Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering

Semester: (Fall, Year:2026), B.Sc. in CSE (Day)

Lab Report : 01
Course Title: Machine LearningLab
Course Code:CSE-404 Section: 241_D3

Lab Report Title : Data Preprocessing Using Python

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Lab Date : 14/08/2026
Submission Date : 24/10/2026
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Lab Report Status

Marks: ……
Signature:.....
Comments:.....
Date:.....

1. Title

Data Preprocessing Using Python

2. Objectives

- Identifying and handling missing values in the dataset.
- Converting categorical data into numerical form using encoding.
- Scaling numerical features using StandardScaler.
- Preparing the dataset for further machine learning tasks.

3. Problem Analysis

Data preprocessing is an important step in machine learning because raw datasets may contain missing values, categorical features, and numerical attributes with different scales. These issues can affect the performance and accuracy of machine learning models.

In this experiment, a synthetic employee dataset was used. The dataset contains employee-related information, including numerical and categorical features. The preprocessing tasks involved identifying missing values, filling missing numerical values with the median and missing categorical values with the mode, converting selected categorical columns into numerical representations using one-hot encoding, and scaling numerical features using StandardScaler.

These operations help make the dataset more consistent and suitable for use in machine learning algorithms.

4. Implementation and Output

The implementation was carried out in Python using the Pandas, NumPy, and Scikit-learn libraries. The code and corresponding outputs for each preprocessing step are uploaded in a separate Jupyter Notebook file.

Implementation and Output: Click here to view the IPYNB file

5. Discussion

In this experiment, the employee dataset was preprocessed through missing-value handling, categorical encoding, and numerical feature scaling. Missing values were replaced using appropriate statistical methods, while one-hot encoding converted selected categorical features into numerical columns. StandardScaler was then applied to the selected numerical features to standardize their scales.

These preprocessing techniques helped improve the consistency and usability of the dataset. The experiment also demonstrated how Python libraries can simplify common data preprocessing tasks before applying machine learning algorithms.

6. Conclusion

In conclusion, the experiment demonstrated the basic steps of data preprocessing using Python. Missing values were handled, categorical features were encoded, and numerical features were standardized. These steps prepared the dataset for further machine learning tasks and provided practical experience with Pandas and Scikit-learn.